

FINAL PROJECT: ROW CROP INTELLIGENCE DATA DASHBOARD

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	9 (Final Project)
TITLE	Row Crop Intelligence Data Dashboard
GRADE VALUE	30 points
DUE DATE	April 7, 2026, 11:59 PM PT

OBJECTIVES

The Final Project is the culmination of your work over the last 8 weeks. You will move beyond static notebooks to build a professional-grade dashboard prototype that simulates a real-world "Row Crop Intelligence" platform.

By completing this project, you will demonstrate the ability to:

- Integrate disparate agricultural data sources (Soil, Weather, Satellite, Yield).
- Build an interactive user interface using Python-based BI tools (like Streamlit or Dash).
- Use AI to generate narrative summaries that explain data trends to non-technical users.
- Present a "Portfolio-Ready" project that showcases end-to-end data engineering and analytics skills.

INSTRUCTIONS

Step 1: Create Your Final Branch

Open your terminal in VS Code:

```
git checkout -b feature/final-project-dashboard
```

Step 2: Dashboard Architecture & Layout

Your dashboard should be organized into logical sections that a farm manager or crop insurance adjuster would find useful.

01. **KPI Summary Tiles:** Create "at-a-glance" cards for your entire 200-field package (e.g., Average NDVI, Total Acreage, Predicted Total Bushels).
02. **Navigation:** Ensure users can filter the view by Field ID or Soil Type.

Step 3: Integrate Your "Best Of" Visualizations

Bring in at least **5 visualizations** developed during Assignments #3–#8. You should refine these to ensure they share a consistent visual theme.

- **Geospatial Map:** From Assignment #4/7 (Fields + Soil/NDVI).
- **Weather Time-Series:** From Assignment #6.
- **Soil Health Metrics:** From Assignment #8.
- **Vegetation Health:** From Assignment #5.
- **EDA Relationship Plot:** From Assignment #3.

Step 4: Adding "Intelligence" with AI Narratives

A dashboard shouldn't just show data; it should provide advice.

01. **Task:** Use your AI assistant to help write the code for "dynamic captions."
02. **Example:** If the average NDVI for a selected field is below 0.3, the dashboard should automatically display a text box saying: *"Alert: Low vegetation index detected. Recommend immediate scouting for nitrogen deficiency or pest pressure."*

Step 5: Final Documentation

A dashboard shouldn't just show data; it should provide advice.

01. **README.md:** Update your main repository README with a project description, a list of technologies used, and instructions on how to run the

dashboard.

- 02. **AI Usage Summary:** Create a short markdown file ([docs/AI_DOCS.md](#)) summarizing how AI helped you overcome technical hurdles during the course.

GRADING CRITERIA (30 POINTS TOTAL)

CRITERION	POINTS	DESCRIPTION
Technical Integration	10	Dashboard successfully connects to and displays data from at least 4 previous assignments.
UI/UX Design	8	Layout is intuitive, visualizations are polished, and the interface is interactive (filters/sliders).
Agronomic Insight	7	The dashboard provides clear, actionable "Intelligence" (KPIs and AI-narratives) for row crop management.
Code Quality	5	Repository is well-structured, documentation is complete, and code follows industry standards.

DELIVERABLES & SUBMISSION INSTRUCTIONS

Part 1: The Deliverables

- 01. **GitHub Repository Link:** The URL to your final branch.
- 02. **Project Demo:** * A recorded screen-walkthrough (2-3 minutes) of your dashboard functionality.

- a. **OR** 3-5 screenshots showing different views/filters of the live dashboard.
- 03. **Final Reflection:** A brief note in the GC comments on your biggest "aha!" moment of the course.

Part 2: Finalize Your Work on GitHub

- 01. **Merge to Main:** Once you are satisfied, merge your feature branch into `main`.
- 02. **Final Push:** `git push origin main`.

Part 3: Submit in Google Classroom

- 01. **Open the Final Project:** Navigate to **Assignment #9** in Google Classroom.
- 02. **Add Your Link & Media:** Attach your GitHub URL and your demo video/screenshots.
- 03. **Turn In:** Click the **Turn In** button.

NOTE ON FEEDBACK: As this is the final project, Clayton will provide a comprehensive review of your end-to-end pipeline. Final grades will be posted alongside your Certificate of Completion.